AIID Research Notebook

User Guide

A Plain-Language Walkthrough

No Programming Knowledge Required

TAI Hackathon — Problem Statement 5

Data: AI Incident Database (CC BY-SA 4.0)

> **How to use this guide:** Open the notebook in Google Colab and run each cell by clicking the **play button** (❖) on the left side of the cell. This guide tells you what each cell does and what to expect after it runs. You do not need to understand the code — just press play and watch the results.

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1 Before You Begin

1.1 What Is This Notebook?

The AI Incident Database (AIID) is a public collection of real-world cases where artificial intelligence caused harm — discrimination, privacy violations, safety failures, misinformation, and more. It currently contains **1,367 documented incidents** spanning from **1983 to 2026**.

This notebook takes that raw data and turns it into **interactive charts, maps, and a searchable web tool** so anyone can explore and understand patterns in AI harms — without writing a single line of code.

1.2 What Do You Need?

- A Google account (to open the notebook in Google Colab)
- An internet connection
- About 5–10 minutes of patience while cells run

1.3 How to Run a Cell

Every gray block in the notebook is a "cell." To run it:

1. Click on the cell to select it.
2. Click the **play button** (❖) on the left side, *or* press **Shift** + **Enter** on your keyboard.
3. Wait for the spinning icon to stop. When it finishes, a green checkmark appears.
4. Move to the next cell and repeat.

> **Shortcut:** To run the entire notebook at once, go to **Runtime** → **Run all** in the top menu bar. Then sit back and watch everything execute from top to bottom.

1.4 Important Setup Step

Before running anything, set your notebook to use a **T4 GPU** (this is free):

1. In the top menu, click **Runtime** → **Change runtime type**
2. Under "Hardware accelerator," select **T4 GPU**
3. Click **Save**

This is needed for the AI-powered chatbot feature in the web application (Section 6). Everything else works without it, but it is best to set it up from the start.

2 Section 0 — Getting Ready

These cells prepare the notebook to work. Think of it like opening an app before you can use it.

2.1 Cell 3 — Install Required Tools

What it does: Downloads the free, open-source software tools that the notebook needs — things like charting tools, data processing tools, and the web application builder. These are standard tools used by data scientists worldwide.

How long it takes: About 30–60 seconds.

> **Look for:** A series of progress bars finishing. No error messages in red.

2.2 Cell 5 — Load the Tools

What it does: Activates all the tools that were just installed, so the notebook can use them.

> **Look for:** The message: `"All imports successful"`

If you see that message, everything is ready.

3 Section 1 — Getting the Data

These cells download the real incident data directly from the official AI Incident Database website. The data is publicly available and freely licensed — anyone can access and verify it.

> **Disclaimer:** Numerical values regarding incident totals and dataset entries are accurate as of February 26, 2026. Due to the dynamic nature of the AI Incident Database, these metrics may fluctuate following subsequent updates or future releases. This is just to demonstrate how things look like on the notebook.

3.1 Cell 8 — Find the Latest Data

What it does: Visits the AIID website and finds the most recent copy of the entire database. This ensures you are always analyzing the freshest available data.

> **Look for:** Three lines showing the snapshot name, date, and download link. For example:
>
> ```
> Latest snapshot:  backup-20260223102103.tar.bz2
> Snapshot date:  2026-02-23
> URL: https://pub-72b2b...
> ```

The date tells you exactly when this copy of the database was created.

3.2 Cell 10 — Download the Data

What it does: Downloads the database file (about 91 megabytes). If you have already run this notebook before, it is smart enough to skip the download and reuse the existing file.

How long it takes: 1–3 minutes depending on your internet speed.

> **Look for:** A progress bar filling up, followed by: `"Download complete:  91.3 MB"`

3.3 Cell 12 — Unpack the Data

What it does: The downloaded file is compressed (like a ZIP file). This cell unpacks it and shows you what is inside.

> **Look for:** A list of files including:
>
> - **incidents.csv** — the main incident records
> - **classifications_MIT.csv** — harm categories assigned by MIT researchers
> - **classifications_GMF.csv** — what the AI was trying to do and how it failed
> - **classifications_CSETv1...csv** — severity and location information
>
> These are the raw files exactly as the AIID publishes them. Nothing has been changed.

4 Section 2 — Loading the Data

These cells read the downloaded files into the notebook so they can be analyzed.

4.1 Cell 15 — Prepare the Data Reader

What it does: Sets up a helper that knows how to correctly read fields in the database. Some entries contain lists (for example, one incident might involve multiple companies), and this helper ensures those lists are read properly.

> **Look for:** No visible output. The cell simply finishes without errors.

4.2 Cell 17 — Set Up the Loading Method

What it does: Prepares the method for reading data files. It first tries the standard spreadsheet format; if that is unavailable, it uses an alternative format. This makes the notebook flexible enough to work with any version of the database.

> **Look for:** No visible output. The cell finishes without errors.

4.3 Cell 19 — Load All Five Data Tables

What it does: Reads all the data into five tables:

Table	What It Contains
Incidents	The core records: 1,367 documented AI incidents, each with a date, title, description, and the organizations involved.
Reports	6,687 news articles, blog posts, and research papers that document those incidents.

MIT Classifications	Harm categories assigned by MIT researchers. Covers about 91% of incidents. Categories include discrimination, privacy violations, safety failures, and more.
GMF Classifications	Records of what the AI system was *trying* to do (its goal) and *how it failed* (the technical failure). Covers about 24% of incidents.
CSET Classifications	Additional details including which country the incident occurred in, whether anyone was injured, and which industry sector was involved. Covers about 36% of incidents.

Look for: Messages confirming each table loaded, for example:

```
incidents from CSV: (1367, 9)
reports from CSV: (6687, 21)
MIT from CSV: (1242, 8)
GMF from CSV: (326, 21)
CSETv1 from CSV: (298, 65)
```

The first number is the row count (how many records). The second is the column count (how many pieces of information per record).

5 Section 3 — Preparing the Data for Analysis

These cells organize and combine the five tables into one complete dataset. This is like merging five separate spreadsheets into a single master spreadsheet, so that every incident has all available information in one place.

What this section does NOT do:

- It does **not** change any incident descriptions
- It does **not** add information that is not in the original database
- It does **not** remove any incidents
- It does **not** re-categorize any incidents

It only standardizes formatting (like making dates consistent) and combines the tables.

5.1 Cell 22 — Company Name Corrections

What it does: Some companies have changed their names over the years. For example, Facebook became Meta. This cell maps about 15 well-known name changes so that incidents involving the same company are counted together rather than split across two names.

Look for: No visible output. The cell finishes without errors.

5.2 Cell 24 — Clean the Incidents Table

What it does: Standardizes column names and converts dates into a format the computer can sort and chart by. Extracts the year from each date for grouping.

> **Look for:** A confirmation showing all 1,367 incidents are preserved and the date range:
>
> ```
> incidents_clean:  (1367, 10)
> Date range:  1983-09-26 -> 2026-02-17
> Null dates:  0
> ```
>
> The row count of 1,367 confirms nothing was removed.

5.3 Cell 26 — Clean the Reports Table

What it does: Standardizes date fields and website names in the 6,687 news reports. No records are removed.

> **Look for:** `reports_df:  (6780, 21)`

5.4 Cell 28 — Calculate Reporting Statistics

What it does: For each incident, calculates how many news articles covered it, when the first and last articles appeared, and how long the media covered it. This information is added to the incidents table.

> **Look for:** No errors. The cell finishes with a brief summary of the merged data.

5.5 Cell 30 — Clean the Classification Tables

What it does: Standardizes the three classification tables (MIT, GMF, CSET). Replaces placeholder values like "–" or "N/A" with proper blanks so they do not accidentally appear in charts as real categories.

> **Look for:** Confirmation messages for each table, for example:
>
> ```
> MIT clean:  (1242, 8)
> GMF clean:  (326, 21)
> CSET clean:  (298, 65)
> ```

5.6 Cell 32 — Build the Master Dataset

What it does: This is the key step. It combines the incidents table with all three classification systems into one unified table. Think of it as taking three expert reports and stapling them to each incident file.

If a classification system did not evaluate a particular incident, that field is simply left blank. No incident is ever removed.

> **Look for:** `incidents_df (master):  (1367, 25)`
>
> Still 1,367 incidents — now with up to 25 columns of information each.

5.7 Cell 34 — Expand Entity Lists

What it does: Some incidents involve multiple companies or multiple harmed groups. This cell expands those lists so each company and each harmed group can be counted individually.

> **Look for:** A data quality summary showing:
>
> - **1,456 unique deployers** (organizations that used the AI)
> - **639 unique developers** (organizations that built the AI)
> - **2,946 unique harmed parties** (people or groups affected)
> - Coverage percentages for each classification system

6 Section 4 — Filters

6.1 Cell 37 — Interactive Filter Panel

What it does: Creates a control panel with sliders and dropdown menus that let you focus on specific parts of the data. Every chart that follows automatically updates based on your selections.

Filter	What It Does
Year Range	Drag the slider to focus on any time period (1983–2026)
Risk Domain	Choose a harm category (e.g., "Discrimination," "Privacy")
AI Goal	Filter by what the AI was trying to do
Deployer	Focus on incidents involving a specific company
Harmed Party	Focus on incidents affecting a specific group

> **Look for:** A row of interactive controls (sliders and dropdown menus) appearing below the cell. Try changing them — the charts in the next section will respond.

7 Section 5 — Visualizations

This is the heart of the notebook. Twelve interactive charts that reveal different dimensions of AI harms across 43 years. Each chart is clickable, hoverable, and zoomable.

7.1 Cell 40 — Chart 1: Incidents Per Year

What it shows: A timeline of how many AI incidents were documented each year, plus the year-over-year growth rate.

What to notice: The line is flat through the early 2000s, begins rising around 2016 (autonomous vehicle incidents), accelerates in 2018 (facial recognition debates), and spikes sharply

from 2022 onward (the rise of ChatGPT and generative AI). Reporting has grown roughly **5× since 2018**.

> **Look for:** A chart with bars showing annual counts and a line showing the growth trend.

7.2 Cell 42 — Chart 2: Harm Categories Over Time

What it shows: A colored area chart showing how different types of AI harm have grown over the years. Each color represents a different harm category from the MIT taxonomy.

What to notice: In the early years, safety and system failures dominate. In recent years, categories like discrimination, misinformation, and privacy violations have grown significantly as a share of total incidents.

> **Look for:** A stacked colored area chart starting from 2010. If the MIT classification data is not available, this cell will display a notice instead of a chart — that is normal.

7.3 Cell 45 — Chart 3: Harm Landscape (Sunburst)

What it shows: A circular, layered chart. The inner ring shows broad harm categories; the outer ring shows specific sub-categories within each. Click any section to zoom in.

What to notice: The largest segment is typically **"AI system safety, failures, and limitations"** — meaning most documented AI harms are accidents and malfunctions, not deliberate attacks. "Discrimination and toxicity" and "Privacy and surveillance" are also prominent.

> **Look for:** A colorful circular chart with two rings. Hover over any section to see its name and count.

7.4 Cell 48 — Chart 4: Top Companies

What it shows: Two side-by-side bar charts: the 20 organizations most frequently listed as *deployers* of AI involved in incidents (left), and the 20 most frequently listed as *developers* (right).

What to notice: A small number of major technology companies appear in a disproportionate share of incidents. This largely reflects their dominant market position — more AI deployment means more potential incidents to be documented.

> **Look for:** Two horizontal bar charts side by side.

7.5 Cell 51 — Chart 5: How AI Goals Fail (Heatmap)

What it shows: A grid where each row is what an AI system was *trying to do* (e.g., recommend content, recognize faces) and each column is *how it failed* (e.g., bias, misidentification). Darker squares mean more incidents at that combination.

What to notice: Certain combinations are hotspots. For instance, content recommendation systems frequently fail by distributing harmful content, while facial recognition systems frequently fail through racial bias and misidentification.

Coverage note: This chart uses data from the GMF classification, which covers about 24% of incidents. It shows a subset, not the complete picture.

> **Look for:** A colored grid with labels on both axes. If GMF data is not available, a notice will appear instead.

7.6 Cell 55 — Chart 6: World Map

What it shows: A world map colored by how many AI incidents were documented in each country.

What to notice: The United States has by far the highest count, followed by other English-speaking countries and parts of Western Europe and Asia. Large areas of the developing world appear blank — this does not mean AI harms do not happen there. It means the documentation is sparse, largely due to English-language media coverage bias.

> **Look for:** A world map with countries shaded in different colors based on incident count.

7.7 Cell 58 — Chart 7: Severity Funnel

What it shows: A funnel that starts wide at the top with all 1,367 incidents and narrows through four stages: all incidents → those with documented tangible harm → those with reported injuries → those with reported deaths.

What to notice: While over a thousand incidents exist, only a small fraction involve physical harm or loss of life. Most AI harms are economic, reputational, or rights-based (discrimination, privacy violations). These non-physical harms affect far more people.

> **Look for:** A funnel-shaped chart with four levels, showing numbers and percentages at each stage.

7.8 Cell 61 — Chart 8: Industry Sectors

What it shows: A bar chart of which industries have the most documented AI incidents — technology, government, healthcare, transportation, finance, and others.

What to notice: The technology sector leads, but AI incidents span a wide range of industries, demonstrating that AI harm is not confined to tech companies alone.

> **Look for:** A horizontal bar chart with industry sector names.

7.9 Cell 64 — Chart 9: Most-Covered Incidents

What it shows: The 15 incidents that received the most media attention, measured by number of news articles written about them. Each bar is colored by the type of harm involved.

What to notice: The most-covered incidents tend to involve well-known companies and easily understood harms (crashes, discrimination lawsuits). Media coverage reflects public interest, not necessarily severity — a quiet data leak affecting millions may receive fewer articles than a viral AI chatbot incident.

> **Look for:** A horizontal bar chart with incident titles on the left.

7.10 Cell 67 — Chart 10: Who Gets Harmed

What it shows: The 20 groups or populations most frequently listed as harmed parties in AI incidents.

What to notice: Broad populations (like "social media users" or "internet users") appear frequently, as do specific vulnerable groups (racial minorities, job applicants). This reveals who bears the brunt of AI failures.

> **Look for:** A horizontal bar chart with harmed party names.

7.11 Cell 70 — Chart 11: Word Cloud

What it shows: A visual display of the most common words found across all 1,367 incident descriptions. Larger words appear more frequently.

What to notice: Prominent words typically include "system," "data," "users," "facial," "recognition," "content," "autonomous," and "vehicle" — reflecting the dominant themes in AI incident documentation over four decades.

> **Look for:** A colorful cloud of words in varying sizes.

7.12 Cell 73 — Chart 12: Shifting Risks

What it shows: A side-by-side comparison of harm categories before 2023 versus from 2023 onward. This reveals how the *types* of AI harm are changing as AI technology evolves.

What to notice: Categories related to misinformation and content generation have grown in their share of incidents since 2023 (the generative AI era), while some historically dominant categories like physical safety have declined proportionally. The nature of AI harm is evolving alongside the technology.

> **Look for:** A grouped bar chart with two colors representing the two time periods.

8 Section 6 — Web Application

These two cells build and launch a fully functional web tool that anyone can use from their browser, even without the notebook open.

8.1 Cell 75 — Prepare the Web App

What it does: Sets up the dropdown options, search functions, and display logic for the web application. This is the behind-the-scenes preparation.

> **Look for:** No visible output. The cell finishes without errors.

8.2 Cell 76 — Launch the Web App

What it does: Starts the web application and gives you a link anyone can visit.

How long it takes: 15–30 seconds.

Look for: A message showing a public link, for example:

```
Running on public URL: https://xxxxx.gradio.live
```

Click this link to open the web application. You can also share it with others — they can use it directly from their browser without needing the notebook.

The web application has five tabs:

Tab	What You Can Do
Incident Explorer	Search for AI incidents by keyword, date range, harm category, or company. Results appear in a sortable table. Click any row to see the full details.
Trend Analysis	Pick any metric from a dropdown (e.g., "Incidents per Year" or "Incidents by Harm Category") and the app generates a chart instantly.
Entity Deep Dive	Select any company or organization from a dropdown. The app shows their complete incident history, a timeline, and a breakdown of what types of harms they have been involved in.
Harm Landscape	Press the "Load" button to see the full harm landscape — a sunburst chart, a world map, and an industry breakdown for the entire dataset.
High Impact	Adjust the slider to see the 10, 20, or 50 most heavily reported incidents with a comparison chart.

Sharing the link: The web application link is active for up to 72 hours. Anyone with the link can access the tool — they do not need a Google account or any software. This makes it easy to share with colleagues, journalists, or policymakers.

9 Section 7 — Honest Limitations

9.1 Cell 78 — Assumptions and Caveats

The final section of the notebook is a written disclosure of what this analysis can and cannot tell you. We include this because responsible data analysis means being upfront about limitations. Here is what you should keep in mind:

DISCLAIMER: It is important to understand that the limitations stated below originate entirely from the **upstream dataset** (the AI Incident Database itself) and not from **design decisions in the notebook code**.

These constraints reflect the current state of the AIID data collection and annotation efforts. Because these are **inherent properties of the AI Incident Database**, they

cannot be resolved by modifying the notebook code or its execution logic.

What You See	What It Actually Means
1,367 incidents	These are *documented and reported* incidents. The true number of AI harms worldwide is certainly higher, as many go unreported or occur in regions without English-language press coverage.
MIT covers 91%	The MIT harm categories are available for 91% of incidents. When you look at charts using these categories, you are seeing nearly the full picture.
GMF covers 24%	The technical failure classifications exist for only about a quarter of incidents. The heatmap (Chart 5) shows a subset, not the full dataset.
CSET covers 36%	Geographic and severity data exist for about a third of incidents. The world map and severity funnel are based on this subset.
Blank areas on the map	These represent missing data, not the absence of AI harm.
Company rankings	A company appearing in many incidents may simply deploy more AI systems. High incident counts correlate with market dominance, not necessarily worse practices.
Media coverage counts	More news articles about an incident reflects media interest, not severity. A quiet data leak affecting millions may get fewer articles than a viral chatbot mishap.

10 Operational Manual: AI Incident Database Explorer

This manual provides a detailed technical guide for navigating the AI Incident Database Explorer to analyze AI-related harms documented since the year 1983 till this date.

10.1 I. Incident Explorer: Primary Data Retrieval

The *Incident Explorer* serves as the central engine for filtering the database's total documented incidents.

1. **Configure Filters:** Input specific terms into the **Keyword Search** (e.g., "deepfake") and refine results using the **Risk Domain**, **Deployer**, or **Harmed Party** dropdown menus.
2. **Temporal Selection:** Adjust the **Year From** and **Year To** sliders to isolate incidents within a specific historical window.
3. **Data Generation:** Click **Search** to populate the results table.

4. **Exporting Data:** Before clicking on "Download", click **Export CSV**. Once the server processes the request, navigate to the *Download* footer and click the filename (e.g., `aiid_filtered_export` to save the file.

10.2 II. Trend Analysis: Longitudinal Visualization

This tab converts raw data into visual trends to identify growth patterns in AI harms.

1. **Select Metric:** Open the **Metric** dropdown to choose a data perspective, such as *Incidents per Year*, *Intent Split*, or *Top 5 Deployers Over Time*.
2. **Render Visualization:** Click the **Generate Chart** button.
3. **Interpretation:** Analyze the resulting line or bar graph. Note the 2-year rolling average line (dark blue) for smoothed trend identification.

10.3 III. Entity Deep Dive: Stakeholder Analysis

Use this tab to investigate the specific footprint of a developer, deployer, or victim.

1. **Set Entity Type:** Choose between **Deployer**, **Developer**, or **Harmed Party**.
2. **Identify Entity:** Use the **Select Entity** searchable dropdown to find a specific organization (e.g., "OpenAI" or "Tesla").
3. **Execute Analysis:** Click **Analyze** to generate a custom dashboard featuring an *Incident Timeline* and a *Harm Type Breakdown* pie chart for that specific entity.

10.4 IV. Harm Landscape: Macro-Systemic Overview

This section provides a high-level view of the taxonomy, geography, and industrial sectors affected.

1. **Initialize Data:** Click the **Load Harm Landscape** button to render global charts.
2. **Taxonomy Exploration:** Interact with the *MIT Risk Taxonomy Sunburst* to see how incidents are classified (e.g., 34% Malicious Actors & Misuse).
3. **Geospatial/Sector Analysis:** Review the *Geographic Distribution* map and the *Top 15 Deployment Sectors* bar chart to identify high-risk regions and industries.

10.5 V. High Impact Incidents: Editorial Significance

This tab ranks incidents by their public visibility and media footprint.

1. **Define Scope:** Adjust the **Number of Incidents to show** slider (range: 10 to 50).
2. **Load Metrics:** Click **Load** to generate the leaderboard.
3. **Cross-Domain Comparison:** Use the *Coverage Chart* to compare media reports across different Risk Domains (color-coded by category).

10.6 VI. AI Assistant: Generative Synthesis

The assistant leverages RAG (Retrieval-Augmented Generation) to answer complex research questions.

1. **Query Submission:** Type a natural language question into the **Your question** field.

2. **Calibrate Retrieval:** Adjust the **Retrieved Incidents** slider to determine how many database records the AI should reference (up to 25).
3. **Verification:** Click **Ask**. To verify the accuracy, toggle **Show retrieved incidents** to view the raw source data used for the answer.

11 Quick Troubleshooting

What Happened	What to Do
A cell shows a red error message	Click **Runtime** → **Restart and run all** in the top menu. This resets everything and runs all cells from the beginning.
A chart says "column not found" instead of showing a chart	This is normal. It means the relevant classification data was not available for that particular chart. Skip it and continue.
The download is very slow	Your internet connection may be slow. Wait for it to finish — it only needs to download once.
The web app link does not work	The link expires after 72 hours. Re-run **Cell 76** to generate a new link.
Charts look empty after changing filters	Your filter combination may be too narrow. Try setting some filters back to "All."

12 Key Takeaways from the Data (As of Feb, 23

After running the full notebook, here are the most important findings:

1. **AI incident reporting has grown dramatically.** The number of documented incidents has increased roughly 5× since 2018, with a sharp spike from 2022 onward coinciding with the rise of generative AI.

2. **Most AI harms are accidents, not attacks.** The single largest harm category is "AI system safety, failures, and limitations" — systems that simply did not work as intended. Intentional misuse exists but is less common.

3. **A small number of companies appear frequently.** The top 20 deployers account for a large share of incidents, mirroring the concentration of AI market share.

4. **Specific AI applications have predictable failure modes.** The heatmap reveals that content recommendation systems tend to fail by spreading harmful content, while facial recognition tends to fail through racial bias.

5. **There is a significant media reporting lag.** On average, more than 30 days pass between an AI incident occurring and the first news article about it.

6. **The nature of AI harm is evolving.** Since 2023, misinformation and content generation harms have grown proportionally, while physical safety incidents have declined as a share of the total.

7. **Geographic data is limited.** The world map primarily reflects where incidents are documented in English-language media, not where AI harms actually occur globally.

13 Data Integrity Statement

This notebook does not generate, fabricate, or modify any data. Every number, category, and label comes directly from the official AI Incident Database. Specifically:

- The data is downloaded from the **official AIID snapshots page**, verified by a date-stamped filename.
- The three classification systems (MIT, GMF, CSET) were created by **independent research teams**, not by the authors of this notebook.
- The only transformations applied are formatting adjustments: standardizing column names, parsing dates, and mapping 15 known company name changes. No incident content is altered.
- Coverage gaps are **explicitly disclosed**: 91%, 24%, and 36% for the three taxonomies respectively.
- The notebook is **fully reproducible**: running it again on the same date produces identical results. Running it on a different date produces results reflecting any new incidents added to the database in the interim.

Data: AI Incident Database — CC BY-SA 4.0.
Built for TAI Hackathon — Problem Statement 5.